

Dr. Tim Hallatt

ASTROPHYSICIST

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Academic Positions

MIT Kavli Institute for Astrophysics and Space Research

POSTDOCTORAL ASSOCIATE

Cambridge, Massachusetts

Sept. 2024 - present

- advisor: Prof. Sarah Millholland

Education

McGill University

PHD, PHYSICS

Montréal, Quebec

Sept. 2021 - July, 2024

- advisor: Prof. Eve J. Lee
- thesis title: "On the Formation of Planets in the Milky Way's Thick Disk"
- topic: theoretical astrophysics
- tools: MESA hydrodynamics/interior structure code, REBOUND dynamics code, Python, Fortran
- additional skills: machine learning with `scikit-learn`

McGill University

MSC, PHYSICS

Montréal, Quebec

Sept. 2019 - Sept. 2021

- advisor: Prof. Eve J. Lee
- thesis title: "Leveraging Exoplanet Occurrence Rates to Test Planet Formation Theory"
- topic: theoretical astrophysics

University of Western Ontario

HON. BSc, PHYSICS

London, Ontario

Sept. 2015 - April, 2019

- honours thesis advisor: Prof. Paul Wiegert
- thesis title: "The Dynamics of Interstellar Asteroids and Comets Within the Galaxy"
- topic: dynamics

Publications

PUBLISHED (FIRST AUTHOR)

- [7] **Hallatt, T.**, Owen, J. E., Millholland, S. 2026. Revealing the Origin of Desert Dwellers via Stellar Obliquities (in press, *ApJL*).
- [6] **Hallatt, T.**, Millholland, S., 2026. Shedding Light on Desert Dwellers. *Astrophysical Journal*, vol. 997, no. 2.
- [5] **Hallatt, T.**, Millholland, S., 2026. Coupled Planetary Interior and Tidal Evolution. *Astrophysical Journal*, vol. 997, no. 2.
- [4] **Hallatt, T.**, Lee, E. J., 2025. On the Formation of Planets in the Milky Way's Thick Disk. *Astrophysical Journal*, vol. 979, no. 120.
- [3] **Hallatt, T.**, Lee, E. J., 2022. Sculpting the sub-Saturn Occurrence Rate via Atmospheric Mass Loss. *Astrophysical Journal*, vol. 924, no. 9.
- [2] **Hallatt, T.**, Lee, E. J., 2020. Can Large-Scale Migration Explain the Giant Planet Occurrence Rate? *Astrophysical Journal*, vol. 904, no. 2.
- [1] **Hallatt, T.**, Wiegert, P., 2020. The Dynamics of Interstellar Asteroids and Comets within the Galaxy: an Assessment of Local Candidate Source Regions for 1I/'Oumuamua and 2I/Borisov. *Astronomical Journal*, vol. 159, no. 4.

STUDENT-LEAD (* - SUPERVISED STUDENT)

- [2] Liveoak, D. *, **Hallatt, T.**, Millholland, S. 2026. Stability of Multiplanet Systems Through Hot Jupiter Destruction. *Astrophysical Journal Letters*, vol. 1002, no. 1.

- [1] Mireles, I., Ulmer-Moll, S., Liveoak, D.* (incl. **Hallatt, T.**) 2026. Uncovering the Rapidly Evolving Orbits of the Dynamic TOI-201 System. [Science Advances](#), vol. 12, issue 16.

NTH-AUTHOR

- [1] Cadieux, C., Plotnykov, M., Doyon, R., et al. (incl. **Hallatt, T.**), 2023. New Mass and Radius Constraints on the LHS 1140 Planets – LHS 1140 b is Either a Temperate Mini-Neptune or a Water World. [Astrophysical Journal Letters](#), vol. 960, no. 1.

WHITE PAPERS

- Benneke, B., Cowan, N., Rowe, J. et al. (incl. **Hallatt, T.**), 2019. Exoplanet instrumentation in the 2020s: Canada's pathway towards searching for life on potentially Earth-like exoplanets. Canadian Long Range Plan for Astronomy and Astrophysics White Papers, LRP2020. [[link to paper](#)]

Seminars & Presentations

- September 2026. [Oral] *Revealing the Origin of Desert Dwellers via Stellar Obliquities*. The Formation, Life and Fate of Exo-Neptunian Systems, Lorentz Center, Leiden, Netherlands (**invited**).

- June 2026. [Poster] *Revealing the Origin of Desert Dwellers via Stellar Obliquities*. Exoplanets 6 conference, Porto, Portugal.

- June 2026. [Oral] *Revealing the Origin of Desert Dwellers via Stellar Obliquities*. AAS Division of Dynamical Astronomy meeting (DDA), Chicago, Illinois.

- December 2025. [Oral] *Shedding Light on Desert Dwellers*. International Conference on Exoplanets and Planet Formation, Tsung Dao Lee Institute (TDLI), Shanghai, China (**invited**).

- November 2025. *Exoplanet Demographics: A Journey Through Space and Time*. Astrophysics Colloquium, Southwest Research Institute (SwRI), Boulder, Colorado (**invited**).

- October 2025. *Shedding Light on Desert Dwellers*. Exoplanet Lunch, Princeton University, Princeton, USA (**invited**).

- June 2025. *Exoplanet Demographics: A Journey Through Space and Time*. Astrophysics Seminar, University of Bern, Bern, Switzerland (**invited**).

- June 2025. *Exoplanet Demographics: A Journey Through Space and Time*. Astrophysics Seminar, Max Planck Institute for Astronomy, Heidelberg, Germany.

- June 2025. *Exoplanet Demographics: A Journey Through Space and Time*. Astrophysics Seminar, Institute of Astronomy, University of Cambridge, Cambridge, UK (**invited**).

- May 2025. *Shedding Light on Desert Dwellers*. [Oral] AAS Division of Dynamical Astronomy meeting (DDA), Atlanta, Georgia.

- April 2025. *Shedding Light on Desert Dwellers*. Exoplanet Lunch, Harvard & Smithsonian Center for Astrophysics, Cambridge, USA.

- October 2024. *On the Formation of Planets in the Milky Way's Thick Disk*. Planet & Star Formation Coffee, Max Planck Institute for Astronomy, Heidelberg, Germany (online; **invited**).

- September 2024. *On the Formation of Planets in the Milky Way's Thick Disk*. [Oral] MIT & Harvard Planetary Meeting, Cambridge, USA.

- June 2024. *On the Formation of Planets in the Milky Way's Thick Disk*. [Poster] Exoplanets 5 conference, Leiden, Netherlands.

- September 2023. *Constraining Planet Formation Theory via Exoplanet Occurrence Rates*. Stars & Planets Seminar, Yale University, New Haven, USA (**invited**).

- July 2023. *On the Formation of Planets in the Milky Way's Thick Disk*. [Oral] Towards Other Earths III: the Planet-Star Connection, Instituto de Astrofísica e Ciências do Espaço, Porto, Portugal

- June 2023. *On the Formation of Planets in the Milky Way's Thick Disk*. [Oral] Emerging Researchers in Exoplanet Science, Yale University, New Haven, USA.

- May 2021. *Sculpting the sub-Saturn Occurrence Rate via Atmospheric Mass Loss*. [Oral] High Energy Exoplanets, European Space Agency XMM-Newton Workshop, Online.

November 2020. *Can Large-Scale Migration Explain the Giant Planet Occurrence Rate?* [Oral] ExoDem Conference, Caltech, Online.

October 2020. *Can Large-Scale Migration Explain the Giant Planet Occurrence Rate?* Exocoffee, Max Planck Institute for Astronomy (online; **invited**).

August 2020. *The Dynamics of Interstellar Asteroids and Comets Within the Galaxy*. [Oral] Division of Dynamical Astronomers Meeting, Online.

June 2020. *The Dynamics of Interstellar Asteroids and Comets Within the Galaxy*. [Poster] American Astronomical Society meeting, Online.

Select Awards & Fellowships

2021	Alexander Graham Bell Canada Graduate Scholarship-Doctoral , NSERC	\$ 105,000
2021	Perseverance Scholarship , McGill University	\$ 1200
2021	L. Trottier Science Accelerator fellowship , McGill University	\$ 5000
2020	Alexander Graham Bell Canada Graduate Scholarship-Master's , NSERC	\$ 17,500
2020	Technologies for Exoplanetary Science Fellowship , NSERC	\$ 6500
2019	Donald R. Hay Prize (for best thesis) , Physics & Astronomy Dept., University of Western Ontario	\$ 300
2019	Dr. Gérard Hébert Scholarship in Physics (for community service, academic excellence, research potential) , Physics & Astronomy Dept., University of Western Ontario	\$ 1700

Media Citations & Interviews

[Science News \(2025\): Galactic chaos at cosmic noon may have stunted Milky Way planet formation](#)

[Astronomy Magazine \(2021\): Our Galaxy's Marvelous Rogues and Misfits](#)

[Scientific American \(2020\): Mystery of Interstellar Visitor 'Oumuamua Gets Trickier](#)

[Nature \(2019\): How Two Intruders From Interstellar Space are Upending Astronomy](#)

[Populär Astronomi \(2019\): Interstellar comet Borisov is a well-known stranger](#)

Mentorship

summer & fall, 2025	DJ Liveoak , Undergraduate/Graduate; primary research mentor for two projects on dynamics of TOI-201 system (Mireles et al. '26), and hot Jupiters with companion planets (Liveoak et al. '26)	MIT
2025	Ritika Sethi , PhD student; research mentorship on coupled orbital/structural evolution of low-mass planets	MIT
fall, 2025	Haedam Im , Undergraduate; research mentorship on stellar rotation periods of hot Neptune hosts	MIT
summer, 2023	Vincent Savignac , Undergraduate; research mentorship on sub-Neptune core-envelope interaction	McGill University
2020-2021	Didar Seghi , Undergraduate; academic non-research mentorship	McGill University
2019-2020	Griffin Schwartz , Undergraduate; academic non-research mentorship	McGill University
2019-2020	Harper Sewalls , Undergraduate; academic non-research mentorship	McGill University

Software

Author of PSAND: [Planet Structure AND Dynamics code package](#) (published in Hallatt & Millholland 2025)

Service & Outreach

2026	Participant , MIT Leadership and Professional Strategies and Skills (LEAPS) Program	MIT
2020-present	Peer reviewer , AAS Journals, Monthly Notices of the Royal Astronomical Society, Astronomy & Astrophysics	
2026	Postdoc Lunch coordinator , MIT Kavli Institute	MIT
September, 2025	Postdoc representative , MIT Physics visiting committee	MIT
August, 2025	Panelist/judge , The Center for Excellence in Education's <i>Research Science Institute (RSI) encore presentations</i> (STEM research program)	MIT
August, 2023	McGill STEM summer camp , Science Discussion/Q+A Group Leader	McGill University
2023	Trottier Space Institute , arXiv paper discussion organizer/leader	McGill University
2020-2022	Trottier Space Institute , meeting with seminar speaker organizer/leader	McGill University
2021-2022	McGill Graduate Association of Physics Students , VP Academic	McGill University
2021-2022	McGill Graduate Association of Physics Students mentorship program , lead organizer	McGill University
2019-2022	McGill Graduate Association of Physics Students mentorship program , mentor	McGill University
2021-2022	McGill Graduate Association of Physics Students , meeting with colloquium speaker organizer/leader	McGill University
2019-2022	McGill Hackathon , mentor	McGill University
2022	Vanderbilt Astronomy Club , public lecture, online	Vanderbilt University
2021	AstroMcGill public lecture. Our Galactic Neighbourhood: Insights From Exoplanets and Interstellar Objects , online	McGill University
2018-2019	Physics and Astronomy Students' Association , President	University of Western Ontario
2016-2019	Physics and Astronomy Students' Association Help Center , lead organizer/tutor	University of Western Ontario